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Paleomagnetism of mid-Cretaceous red beds in west-central Kyushu Island, southwest Japan: paleoposition of Cretaceous sedimentary basins along the eastern margin of Asia

Kazuto Kodama [a,*], Tetsuichi Takeda [b]

[a] *Marine Core Research Center, Kochi University, Kochi 780-8520, Japan*
[b] *Metal Mining Agency of Japan, Tokyo 105-0001, Japan*

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Abstract

A paleomagnetic study was carried out on the mid-Cretaceous sedimentary strata in west-central Kyushu Island, southwest Japan, to elucidate the origin of sedimentary basins along the Asian continental margin in the Cretaceous. We collected paleomagnetic samples from a total of 34 sites of the mid-Cretaceous Goshonoura Group, shallow-marine clastic deposits in west-central Kyushu, and characteristic remanent magnetizations were recognized from 18 horizons of red beds. Thermal demagnetization has revealed that the red beds contain three magnetization components, with low (<240°C), intermediate (240–480°C), and high (480–680°C) unblocking temperatures. The low unblocking temperature component is present-field viscous magnetization, and the intermediate one is interpreted as chemical remanent magnetization carried by maghemite that was presumably formed by post-folding, partial oxidation of detrital magnetite. Rock magnetic and petrographic studies suggest that the high unblocking temperature component resides largely in hematite (martite and pigmentary hematite) and partly in maghemite. Because of the positive fold test, this high temperature component can be regarded as primary, detrital remanent magnetization. The tilt-corrected mean direction of the high temperature component is Dec = 65°, Inc = 63° with $\alpha_{95} = 5°$, which yields a paleomagnetic pole at 39°N, 186°E and $A_{95} = 8°$. A combination of this pole with those of the Late Cretaceous rocks in southwest Japan defines an apparent polar wander path (APWP), which is featured by a cusp between the Late Cretaceous and the Paleogene. A comparison of this APWP with the coeval paleomagnetic pole from northeast Asia suggests an approximately 50° post-Cretaceous clockwise rotation and $18 \pm 8°$ southward drift with respect to northeast Asia. The southward transport of the Cretaceous basin suggests that the proto-Japanese arc originated north of its present position. We propose that the coast-parallel translation of this landmass was caused by dextral motion of strike-slip faults, which previous geodynamic models interpreted to be sinistral through the Mesozoic. The change in strike-slip motion may have resulted from Mesozoic collision and penetration of exotic terranes, such as the Okhotsk microcontinent, with the northeastern part of Asia. © 2002 Elsevier Science B.V. All rights reserved.

Keywords: paleomagnetism; Cretaceous; red beds; Kyushu; continental drift

* Corresponding author. Tel.: +81-888-44-8325; Fax: +81-888-44-8356.
E-mail address: kdma@cc.kochi-u.ac.jp (K. Kodama).

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1. Introduction

It is widely believed that the Cenozoic tectonics of East Asia was influenced to a large extent by the India–Asia collision [1]. However, the late Mesozoic tectonic evolution of East Asia, after the amalgamation of major tectonic blocks in East China in the Jurassic [2,3], remains uncertain. Based on paleomagnetism [2] and global plate reconstructions [4], some authors claim that the Eurasian continent has remained stable through the Cretaceous, while others propose that active tectonism, including collision, accretion and vertical-axis rotation of various-sized terranes occurred for the entire Mesozoic in the periphery of East Asia [5,6].

The aim of this study is to present paleomagnetic data from amongst the mid-Cretaceous sedimentary basins in southwest Japan. These Cretaceous sedimentary basins are regarded as pull-apart basins, which were once formed in close relation to strike-slip faults along the Asian continental margin [7]. As the motion of these strike-slip faults is ultimately related to plate interactions and associated intra-continental deformation, determining the paleoposition of these pull-apart basins is expected to be critical for clarifying the tectonic evolution of East Asia during the Cretaceous.

An increasing number of paleomagnetic and geochronological studies of the Tertiary rocks in the Japanese arc [8–10] have clearly documented that the arc experienced rapid clockwise (southwest Japan) and counter-clockwise (northeast Japan), vertical-axis rotations in the Middle Miocene. However, the paucity of paleomagnetic studies of the pre-Tertiary rocks makes it difficult to delineate the origin and development of the proto-Japanese arc prior to the back-arc basin opening. The available Cretaceous data are derived mainly from igneous rocks [9], and those from sedimentary strata, which pass the fold test under well-defined structural and chronostratigraphic control, are scarce. This situation is mainly due to the limited occurrence of suitable outcrops and partly to the early Tertiary remagnetization [11]. A few successful results were obtained from the Late Cretaceous deposits along the Median Tectonic Line (MTL) of southwest Japan [12], from which several Late Cretaceous polarity chrons, as well as their mean geomagnetic pole, were determined. In the quest for a more reliable, pre-Tertiary apparent polar wander path (APWP) of the Japanese arc, we aimed at the Cretaceous deposits distributed along the western extension of the MTL in central Kyushu Island (Fig. 1). We provide the paleomagnetic results from the Goshonoura Island in the Amakusa district of west-central Kyushu, where we could examine Cretaceous, red bed-bearing, marine successions that rarely occur along the Asian continental margin.

2. Geological setting

On Kyushu Island, pre-Cenozoic geologic units trend ENE–WSW and represent part of a Cretaceous to Paleogene accretionary complex [13]. Along the north side of the MTL several Cretaceous sedimentary basins occur as E–W elongated depressions filled with thick clastic deposits of shallow- to deep-sea, partly paralic, facies [7]. They tend to be younger to the east, with the Goshonoura basin of western Kyushu being mid-Cretaceous, while on Shikoku Island the easternmost Izumi basin is latest Cretaceous. The structural and lithological characteristics suggest that they are pull-apart basins caused by Cretaceous strike-slip displacement of the proto-MTL [14]. These marine successions are unique when compared with the coeval deposits on the Asian continent that are mainly terrestrial [15]. Cretaceous red beds of terrestrial origin occur in northern Kyushu and westernmost Honshu Island as well as on the Korean Peninsula. However, many of them are remagnetized [16] and the few pre-folding magnetizations reported so far are from limited localities in the southeastern Korean Peninsula [6,17] and eastern China [2]. This is why we have chosen the Goshonoura Group as a suitable target for a comparative study on the origin and development of Cretaceous basins on the Asian continent and its eastern margins.

The mid-Cretaceous Goshonoura Group crops out in the Goshonoura Island and nearby islets of

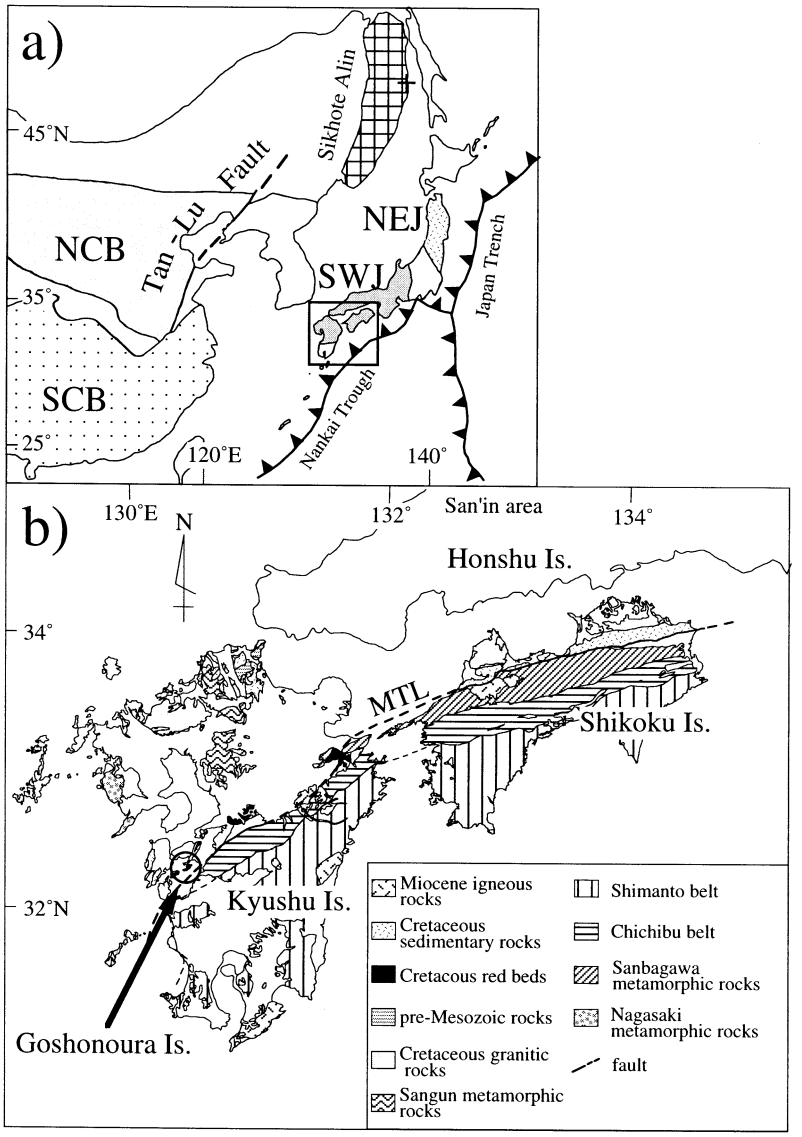


Fig. 1. (a) General tectonic map of northeast Asia, and (b) geologic map of southeast Japan with the location of study area, Goshonoura Island, in west-central Kyushu. NCB = North China Block, SCB = South China Block, SWJ = southwest Japan, NEJ = northeast Japan, MTL = Median Tectonic Line.

the Amakusa area, west-central Kyushu (Fig. 2a). The group is more than 700 m thick and strikes N20°E to 60°E and dips 20–60°, forming a syncline with a NE–SW-trending axis that plunges gently southwest ($<10°$). The Goshonoura Group lies nonconformably on the Maeshima Granodiorite (130–115 Ma) of the Higo metamorphic belt [18,19], and is in fault contact with the overlying, Late Cretaceous Himenoura Group on the western part of the island.

The Goshonoura Group is divided upward into the Eboshi, Enokuchi and Karakizaki formations [18], each of which is generally composed of alternating beds of conglomerates, vari-colored arkosic sandstones and mudstones. Each formation contains slightly different depositional facies

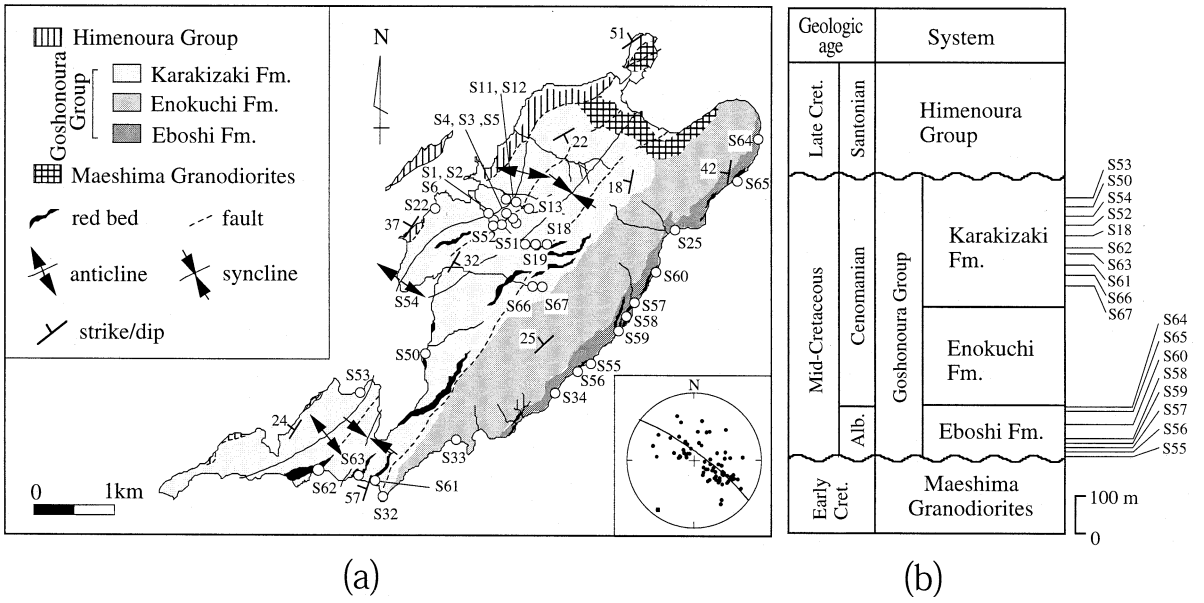


Fig. 2. (a) Geological map of Goshonoura Island (simplified in part after [17]). Sampling sites are indicated by open circles with site indices as S**. Inset, equal-area plots of poles to bedding at 80 localities. (b) Columnar section of the Cretaceous system of Goshonoura Island, showing stratigraphic levels of 18 sites of red beds.

from the other, although collectively they represent estuarine to inner-shelf environments [20]. The Enokuchi Formation is rich in late Albian to early Cenomanian ammonites and bivalves [21,22]. Red, medium- to fine-grained sandstones and siltstones occur in the basal horizons of the Eboshi and Karakizaki formations. Sedimentological and sequence stratigraphic analysis for the group [20] has led to the recognition of four depositional sequences that are in accord with global eustatic sea-level change around the Albian/Cenomanian boundary.

3. Paleomagnetism

3.1. Field and laboratory procedures

We collected a total of 376 samples at 34 sites of the Eboshi, Enokuchi, and Karakizaki formations, including 18 sites (143 specimens) of red beds (Fig. 2b). At each site, more than six cores of 1″ in diameter were bored by a portable drill and oriented with a magnetic compass, and a sun compass for correcting declination anomalies. Magnetic measurements and demagnetizations were carried out in a magnetically shielded room at Kochi University, using either a 2G Enterprises three-axis cryogenic magnetometer or a Schonstedt spinner magnetometer and laboratory-made thermal and alternating-field (AF) demagnetizers.

A pilot study was carried out for two specimens per site, each being subjected to progressive thermal and AF demagnetization to determine proper demagnetization for the remaining samples. Thermal demagnetization was made with temperature steps ranging from 160° to 680°C with 30–40°C intervals, and AF demagnetization used steps from 5 mT to 80 mT with 2.5–10 mT increments. According to the pilot measurements, the remainder of the samples were demagnetized, either thermally or magnetically, using more than 10 steps per specimen. After each thermal demagnetization run, low-field bulk susceptibility was measured with a Bartington susceptibility meter to detect potential alteration of magnetic minerals. The characteristic remanent magnetization (ChRM) component was determined applying principal component analysis [23] to a vector segment con-

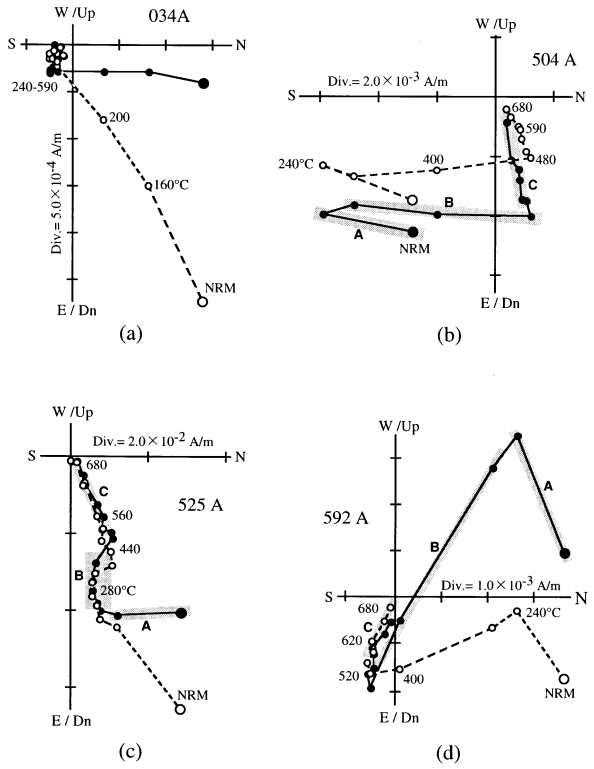


Fig. 3. Examples of vector demagnetization diagrams: (a) grayish siltstone with a single low-coercivity magnetization component, (b–d) red beds with three ChRM components, demagnetization steps for each being indicated by shaded areas labeled A, B and C. (a,c) Examples of detailed demagnetization for the pilot samples. Solid (open) circles are projections onto horizontal (vertical) plane in geographic coordinates.

sisting of at least three end points. A site mean direction was calculated using Fisher statistics [24], and those having a 95% confidence radius smaller than 20° were regarded as acceptable to calculate an overall site mean direction.

Bedding-tilt correction was performed taking the plunging syncline into consideration. The orientation of the plunging fold axis was deduced from the best-fit great circle that was defined with plotting poles to bedding measured at a total of 80 different localities. The mean fold axis obtained this way has a trend of 36° E and a plunge of 10° SW. Consequently, tilt corrections consisting of two successive procedures were carried out; first, rotating the in situ magnetic directions and bedding pole directions by the plunge of the fold (10°) about a horizontal axis normal to the fold axis, and second, restoring the respective magnetic directions through the dip of the reoriented bed about the new strike direction as in ordinary tilt corrections.

Thermomagnetic analysis using a Curie balance was carried out for selected specimens in both air and vacuum. The heating and cooling rate was 8°C/min with applying a direct field of 0.7 T. Also, acquisition of isothermal remanent magnetization (IRM) in stepwise fashion up to 2 T was made with using an impulse magnetizer. Thermal demagnetizations of three composite IRMs [25], each being imparted orthogonally with fields of 0.12 T, 0.4 T and 2 T, were carried out with temperature steps from 80°C to 700°C with 30–40°C intervals.

3.2. Results

Natural remanent magnetizations (NRMs) of gray to brownish sedimentary rocks range in in-

tensity from 5×10^{-4} to 2×10^{-3} A/m, and are rapidly demagnetized during a few early steps, lower than 10 mT or 200°C, above which no meaningful components are recognizable (Fig. 3a). The single ChRM components are almost parallel to the present geomagnetic field, and they are regarded as a recent overprint, or viscous remanent magnetization (VRM); thus they were omitted from further analysis.

In contrast, the red beds have stronger NRM, with initial intensities of 7×10^{-3} to 8×10^{-2} A/m, which are demagnetized effectively by thermal demagnetization, and generally consist of three ChRM components, labeled A, B, and C (Fig. 3b–d). The first component, component A, is recognizable from about 90% ($n=134$) of red bed specimens, and removable with temperature steps up to 240°C, or AF steps below 20 mT. The overall mean direction, before tilt corrections, is Dec = 13°, Inc = 44°, $\alpha_{95}=5°$, and $\kappa=6.8$. Because of the close parallelism to the dipole field (Inc = 51°) and no improvement in directional scatter by tilt corrections, we conclude that component A is post-folding VRM as well as the single component of gray sandstones and siltstones.

Component B has moderate unblocking temperatures between 240°C and 440°C, and can be recognized for almost all the red bed sites, while there are a few cases in which component B cannot clearly be isolated due to overlap of demagnetization spectra (Fig. 3c). We then applied the method of Hoffman and Day [26], but it turned out that the number of those indirect data was too small (only three) to affect the overall statistics. It is notable that component B from any single site exhibits neither consistent directional grouping nor geomagnetic polarity. Their directions, when plotted on an equal-area net, show large azimuthal scatter lying along a great circle, whose pole is oriented almost vertical in geographic coordinates and inclines moderately in stratigraphic coordinates (Fig. 4). Angular deviation from the best-fit great circle [23] for the latter is 21°, which is slightly greater than for the former (16°). These directional behaviors suggest that component B is unlikely to be remagnetization acquired by a single, uniform thermal event, but rather seems to be related to magnetic mineralogical alteration caused by long-term, diagenetic processes. Though it might be appropriate to call the unusual component B an 'intermediate stability overprint', we will hereafter use 'component B' for the sake of simplicity.

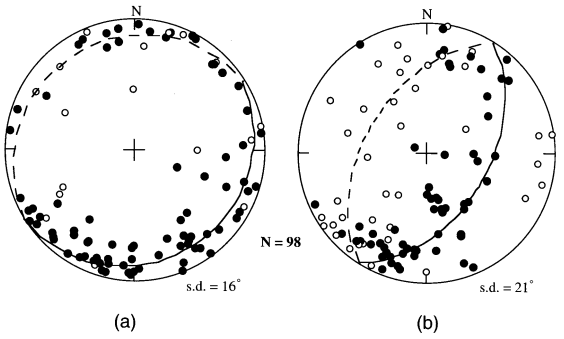


Fig. 4. Directional distributions and their best-fit great circles for component B isolated by least-squares line fitting. (a) In situ (geographic coordinate), (b) unfolded (stratigraphic coordinate). Equal-area projection with solid symbols on lower hemisphere, and open or dashed ones on upper hemisphere. s.d. = angular standard deviation of least-squares fitting.

Component C initially appears around temperature steps of 440–480°C, above which it shows almost linear decay toward the origin. Numbers of components B and C for each site are summarized in Table 1. The unblocking temperature spectrum of component C is unique in that there appear in many cases to be two peaks: one over 560–590°C heating intervals and the other over 650–680°C, suggesting that component C resides not only in hematite. Discussion on the possible carriers of component C, as well as component B, will be given below in association with rock magnetic analyses. The in situ, overall mean direction of the 18 red bed sites (Dec = 90°, Inc = 33°, $\alpha_{95}=9°$, $\kappa=17$) is significantly different from the geocentric dipole field and, after bedding-tilt corrections, shifts more easterly and downwards, with much less directional dispersion, that is Dec = 65°, Inc = 63°, $\alpha_{95}=5°$ with $\kappa=42$ (Fig. 5, Table 1). The improvement in grouping is proven to be statistically significant using a conservative fold test of McElhinny [27], resulting in the ratio of κ indices after and before unfolding, or 42/17 = 2.41, which is greater than the 95% confidence limit of 1.77. The positive fold test suggests that only component C is pre-folding.

Table 1
Paleomagnetic results for component C of red beds from the mid-Cretaceous Goshonoura Group

Site	$N/N_b/N_c$	Str/Dip	In situ		Tilt-corrected		α_{95}	κ	VGP	
			D	I	D	I			Long. (°E)	Lat. (°N)
S18	7/0/7	60E/30W	94.4	44.5	57.4	53.9	3.9	240	201.2	42.7
S50	8/7/8	15W/45S	74.6	15.5	71.7	60.5	6.1	82	189.0	33.6
S52	8/1/7	65W/20W	69.0	48.9	93.1	60.4	8.9	47	182.6	18.6
S53	8/8/7	75E/15S	49.0	48.0	67.4	52.5	5.7	114	199.8	34.3
S54	8/6/6	09W/62W	55.8	12.5	15.2	61.8	8.1	69	174.1	74.0
S55	8/8/7	40E/36W	117.8	23.3	105.3	58.0	13.1	22	180.1	9.2
S56	8/3/7	41E/40W	113.0	27.4	89.3	63.5	5.5	119	180.2	22.7
S57	8/8/6	25E/40W	82.8	33.4	44.3	61.1	6.2	117	191.6	53.8
S58	8/6/6	25E/40W	89.5	22.4	66.2	55.9	8.2	68	196.3	36.3
S59	8/6/7	08E/30W	98.8	32.8	97.1	62.8	4.9	155	178.5	17.3
S60	8/8/7	07E/23W	79.2	44.9	63.1	66.0	6.7	81	182.0	40.8
S61	8/8/6	22E/55W	77.5	13.5	43.0	52.2	7.5	80	207.6	54.0
S62	8/6/6	25E/38W	108.4	23.5	99.1	61.1	4.9	185	179.7	15.1
S63	8/6/6	30E/70W	97.8	8.7	47.7	65.7	10.4	43	182.3	51.1
S64	8/6/7	30E/50W	99.6	20.1	68.1	63.5	5.0	148	185.4	37.0
S65	8/6/7	27E/50W	90.4	31.0	36.8	66.7	5.4	124	177.8	57.9
S66	8/4/7	23E/38W	86.4	40.2	42.0	68.3	3.4	322	175.6	54.0
S67	8/4/6	23E/38W	84.5	35.7	47.3	64.1	8.0	72	185.8	51.5
Site Mean			88.0	30.4	64.7	63.1	5.4	42	186.1	39.4
									(A_{95} = 8.2°, K = 39)	

$N/N_b/N_c$, number of samples studied/number of samples with component B/number of samples with component C; Str/Dip, mean azimuth/dip angle; D and I, declination and inclination in degrees; α_{95}, radii of 95% confidence cone; κ, precision parameter of Fisher statistics; Long. and Lat., longitude and latitude of virtual geomagnetic pole (VGP); A_{95} and K, radius of 95% confidence cone and precision parameter of VGP.

4. Rock magnetism

4.1. Thermomagnetic analysis

Representative examples of thermomagnetic analyses for red bed specimens are shown in Fig. 6. The curve for the first run in air (Fig. 6a) shows about 50% decrease in relative magnetization intensity on completion of cooling, while above 550–580°C it turns out to be reversible showing a Curie temperature (T_c) of hematite around 680°C. The curve of the second run for the same sample (Fig. 6c) appears to be reversible over the almost entire temperature range, showing the hematite T_c around 680°C, as well as another T_c around 550–580°C observable in the first run in air. These curves suggest the presence of maghemite (γFe_2O_3) and its inversion, or oxidation, to hematite (αFe_2O_3) at 550–580°C [28], as well as a small amount of Ti-poor titanomagnetite, or magnetite (Fe_2O_3), as suggested by the inflection repeatedly observed around 550°C. The curve in vacuum (Fig. 6b) is also irreversible between room temperature and around 580°C, but exhibits a more than twofold increase after the cooling

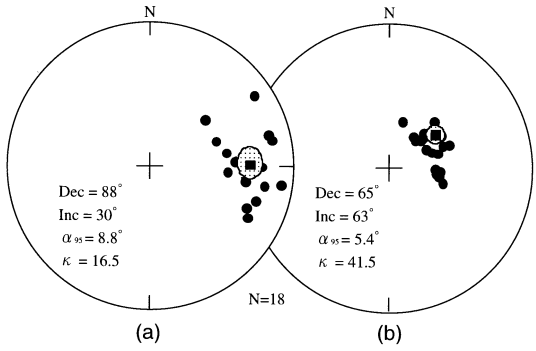


Fig. 5. The site mean directions of component C from red beds, before (a) and after (b) bedding-tilt corrections. The overall mean direction is shown by a square, and the associated 95% confidence limit by an oval. Equal-area projections on lower hemisphere.

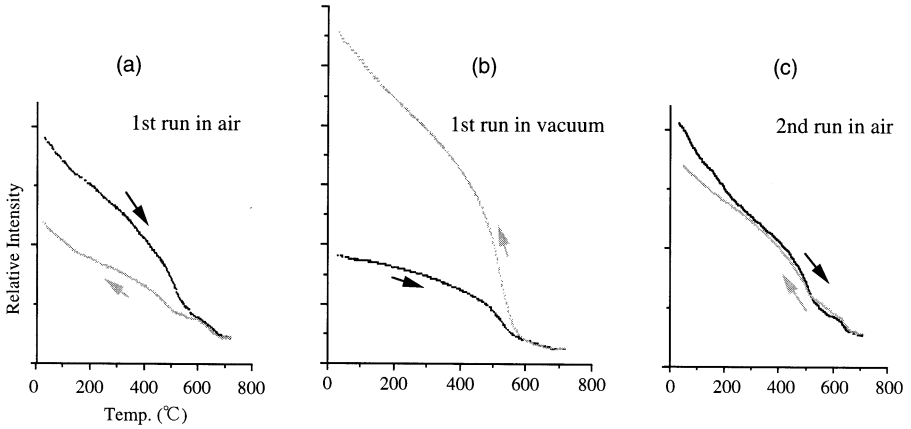


Fig. 6. An example of thermomagnetic analysis for a red bed sample at site S60. (a) First run in air, (b) in vacuum. (c) Second run in air.

run. The drastic increase on cooling down from 580°C appears to be due to magnetite formed by reduction of maghemite over 550–580°C.

These curves suggest that red beds generally contain at least three intrinsic magnetic minerals, such as maghemite, hematite and a small amount of magnetite. The small inflection around 150–200°C may be indicative of oxidation of iron hydroxides such as goethite, or iron sulfides such as pyrrhotite and greigite.

4.2. IRM acquisition and demagnetization

A stepwise IRM acquisition experiment was carried out for those red bed samples which hold all three components (Fig. 7b,c) and, for comparison, that for which component B could not be isolated (Fig. 7a). They appear to be different in acquisition rate for the early steps, with Fig. 7a showing a steep rise at first whereas Fig. 7b,c display a moderate build-up. In excess of 0.3–0.5 T, however, all the IRMs keep increasing with no saturation even at 2 T. The acquisition curves, coupled with corresponding coercivity spectra, suggest the presence of hematite and lower coercivity minerals such as magnetite or maghemite.

Features in accord with the IRM acquisition curves are demonstrated in a different fashion by composite IRM acquisition and thermal demagnetizations (Fig. 8), which were made for each of the three daughter specimens in Fig. 7. It is obvious that the hard (2.0 T) and medium (0.4 T) fractions remain up to 680°C, while the soft fraction (0.4 T) is removed, though not completely, by 580°C. An inflection at around 180–200°C, commonly recognized for the soft fraction and only slightly for the medium fraction, seems to correspond to that observed for the thermomagnetic curves, suggesting dehydration of some iron hydroxides, or presence of small amount of Ti-rich titanomagnetite. We thus consider that this soft IRM fraction corresponds to component A of NRM, which most likely resides in titanomagnetite and/or iron hydroxides. Also, a small inflection of the soft fraction in the 320–400°C range would imply a minor contribution of pyrrhotite.

4.3. Origin of ChRM

Results of the thermal demagnetizations and rock magnetic studies above demonstrate that the major magnetic minerals responsible for ChRMs include not only fine-grained hematite but some others with lower coercivity and unblocking temperature. Direct evidence for this conclusion is provided by the common observation of coarse-grained martite (Fig. 9), which is hematite as a pseudomorph of magnetite. Pigmentary hematite surrounding martite grains is obviously a major carrier of component C, but martite

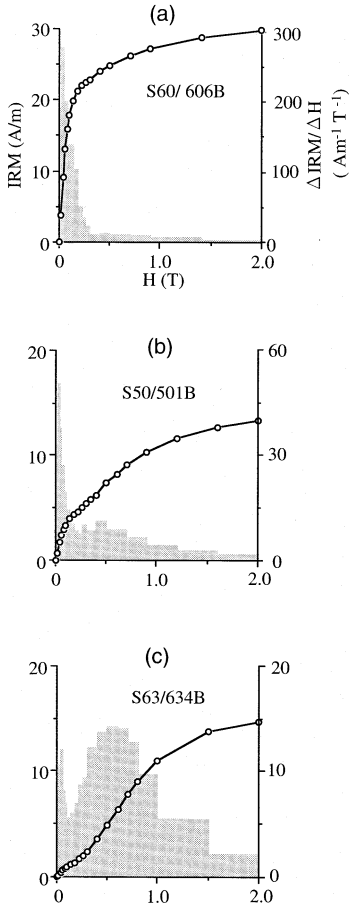


Fig. 7. Three different patterns of progressive IRM acquisition curves and coercivity spectra from red bed samples. (a) Sample 606B at site S60, (b) 501B at S50, and (c) 634B at S63.

is thought to be another potential carrier. Careful observation of the martite grains does not necessarily show homogeneous, complete hematization of magnetite, but rather suggests that there remain small portions consisting of maghemite as a precursor to hematite. The intergrown, fine-grained maghemite could hold a fraction of component C, presumably that with relatively lower unblocking temperature of 400–480°C to 580°C.

There is one observation in the field that seems critical to distinguish the origin of ChRMs of red beds. That is a mixed interface between one red bed and an overlying grayish sandstone layer, which provides evidence of considerable bioturbation (Fig. 10). Commonly, gray burrows penetrate the red bed, yielding a clear contrast in color and lithofacies. These syndepositional features suggest that hematite is one of the primary constituents of red beds. This field observation itself does not preclude the possibility of magnetic overprint followed by the acquisition of components A and B. However, coupled with the lithological and mineralogical contrast between red beds and the overlying gray layers, we view component C as a depositional, or post-depositional, remanent magnetization (DRM, or pDRM). Iron hydroxides, such as goethite, which generally are weathering products in soils, were probably deposited and acquired DRM. We assume that detrital magnetite, which presumably coexisted as a parent of martite at the time of deposition, also gave rise to DRM. Post-depositional maghemitization of the detrital magnetite may have produced chemical remanent magnetization (CRM), which could be responsible for lower unblocking temperature fractions of component C and most of component B as well.

The anomalous, azimuthal scatter of component B as in Fig. 4 emerges in a more intriguing manner upon converting to virtual geomagnetic pole (VGP). Namely, VGPs determined for all the in situ directions of component B also lie along a great circle (Fig. 11). Assuming that component B is CRM of daughter maghemite, then the best-fit great circle, on which a mean VGP of component C lies as well, would help us to assess when maghemitization took place. This is because experiments [29] show that there are CRMs with intermediate directions between the applied field and the parent remanence direction, and they tend to deflect away from the parent remanence as oxidation proceeds. Accordingly, we postulate that the large azimuthal scatter of component B is attributable to inhomogeneous maghemitization and grain size of parent magnetite, and hence the VGPs corresponding to various degrees of maghemitization come to lie on the great circle. Considering that component C has one single normal polarity, the great circle distribution of VGPs of component B may further suggest that the maghemitization occurred over a time interval containing at least one polarity transition, because the vector combination of the remagnetized direc-

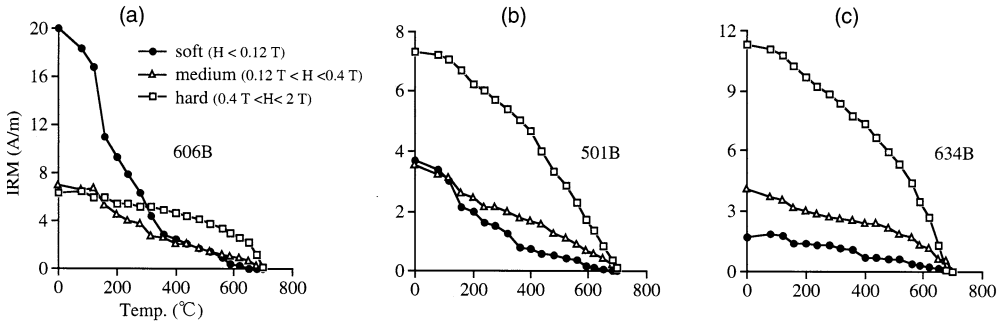


Fig. 8. Results of thermal demagnetizations of three composite IRMs imparted to the same specimens as in Fig. 7. IRMs were imparted with 0.12 T, 0.4 T and 2.0 T of DC field for three orthogonal axes of each specimen.

tion and the primary component C could lie almost fully along one great circle, if the inhomogeneous maghemitization took place according to a rapid movement of the transient pole along one great circle.

5. Discussion

In Fig. 12a we show the overall mean geomagnetic pole of component C from Table 1, together with Cretaceous to Tertiary poles from southwest Japan [30], a mean Cretaceous pole from the stable part of China [2], and that from eastern China [31]. When compiling the poles, we omit data from northeast Japan [32], which suggest a post-Cretaceous, large tectonic rotation quite different from elsewhere. We also exclude results from Korea [6,17] because they are indistinguishable from the mean pole of China. It is apparent in Fig. 12a that paleomagnetic poles from southwest Japan exhibit a unique APWP consisting of two, Cretaceous and Tertiary, paths with different orientations. The two paths appear to intersect forming a cusp between the latest Cretaceous and the Paleogene. Greater deviation of the Tertiary poles with age away from the North pole is attributed to a rapid, clockwise rotation of southwest Japan with respect to Eurasia [10], but the Cretaceous path based on results from sedimentary rocks is drawn for the first time by this study.

Since the amalgamation between the North China Block (NCB) and the South China Block (SCB) in the Jurassic [2,3], there appears to have been no large internal deformation in the eastern Asian continent. It is claimed that collision of the

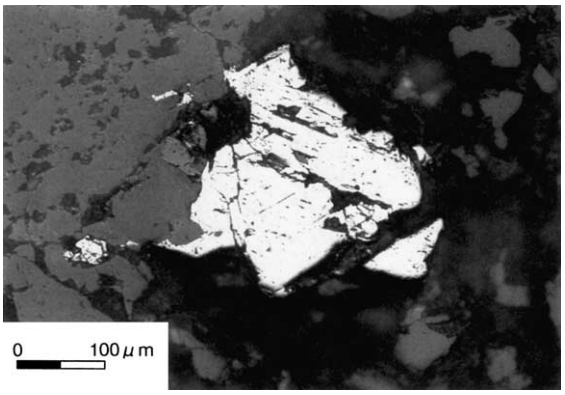


Fig. 9. Photomicrograph under reflected light, showing the occurrence of a large (~100 μm) martite grain (light gray). Sample 564B at site S66.

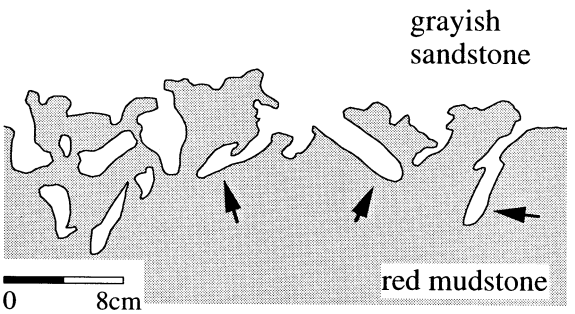


Fig. 10. A close sketch of a mixed interface between red mudstone and overlying grayish sandstone outcropping at the lower part of the Eboshi Formation of the Goshonoura Group. Arrows show sand pipes burrowing into the red mudstone layer.

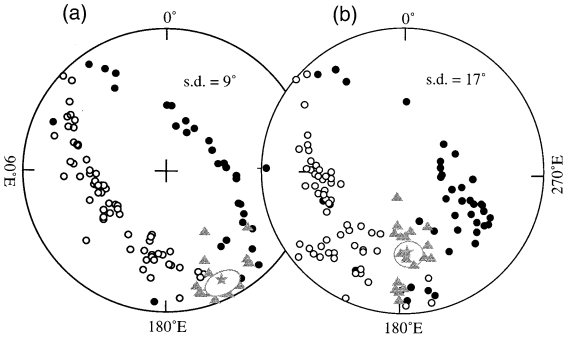


Fig. 11. VGPs of component B (circles) calculated from the directions of red bed samples shown in Fig. 4, (a) before tilt corrections, (b) after tilt corrections. VGPs of component C (gray triangles), their mean position (gray star) and 95% confidence limit (gray oval) are also shown. Solid and gray symbols are equal-area projections on the northern hemisphere, and open ones on the southern hemisphere. s.d. = angular standard deviation of component B when fitted to a great circle by least-squares calculation.

Indian plate with Asia in the early Tertiary gave rise to lithospheric deformation, which eventually led to southward extrusion of Indochina in excess of 10° as well as clockwise rotation of tens of degrees [33,34]. However, to the north, the coeval tectonic reactivation of the united NCB–SCB block appears to be minor and the paleomagnetic directional anomaly expected from the Cenozoic Eurasian APWP is smaller than 10° [4]. This is supported by the combined Cretaceous paleomagnetic pole from NCB and SCB, which is located close to the North pole (pole 8 in Fig. 12a). There is another Cretaceous paleomagnetic pole showing more easterly deviation (7), which is based on data from the Cretaceous rocks to the east of the Tan-Lu fault system, one of the major tectonic lineaments running in a NNE–SSW direction in the eastern NCB and Mongolia. This Cretaceous pole can be regarded as exceptional and may represent rotations of detached blocks [31]. However, we would like to point out that the Cretaceous portion of the hairpin-curved APWP of southwest Japan is distinguishable from any of the contemporaneous poles of Eurasia.

A comparison of the mid-Cretaceous pole of this study (1 in Fig. 12a) with that of the united China block (8) yields a clockwise, vertical-axis rotation by $51 \pm 11°$ with respect to the stable China, and another comparison with the 'anomalous' pole (7) suggests a smaller clockwise rotation by $37 \pm 10°$. These rotations are most likely ascribable, as already claimed by many studies, to the opening of the Japan Sea in the Miocene. To highlight the difference among these Cretaceous poles, we made an attempt to remove the effect of the Miocene rotation by back-rotating the Cretaceous path for southwest Japan about an Euler pole for rotation at 129°E, 34°N, which was determined by paleomagnetism of Miocene volcanic rocks [35]. This results in an APWP wherein the pole from the China block lies midway along the Cretaceous path for southwest Japan (Fig. 12b), with a slight overlap of 95% error circles. The Cretaceous path for southwest Japan as a whole shows a latitudinal translation with respect to the China block over Cretaceous time. Because the pole of the united China block, after docking to Eurasia in the late Jurassic, came to a standstill

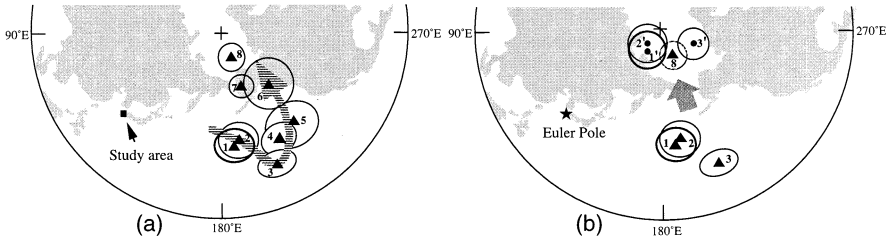


Fig. 12. (a) Cretaceous to Paleogene paleomagnetic poles from northeast Asia (triangles), including southwest Japan (1–6), east of the Tan-Lu fault in China (7) and NCB (8). 1, Mid-Cretaceous, component C of this study; 2, mid-Cretaceous [9]; 3, mid- to Late Cretaceous [30]; 4, Paleogene [9]; 5, Miocene [9]; 6, Oligocene [42]; 7, Cretaceous [31]; 8, Cretaceous [2]. A tight hairpin curve shows the southwest Japan APWP. (b) Comparison of the Cretaceous paleomagnetic poles of southwest Japan (1–3) to those with the effect of the Miocene clockwise rotation removed (1′–3′) and the NCB pole (8). The Euler pole for the Miocene rotation at 129°E, 34°N [35].

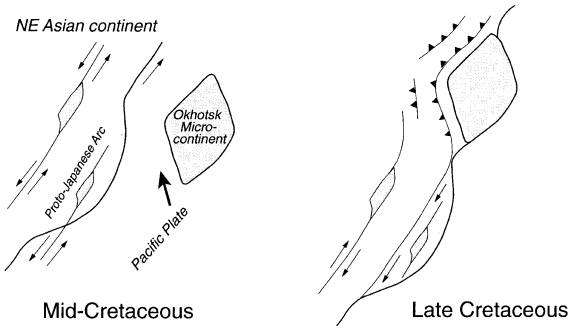


Fig. 13. A simplified tectonic model for the northeastern margin of Asia over the mid- to Late Cretaceous, showing the southward extrusion of the proto-Japanese Arc caused by the collision of the Okhotsk Sea microcontinent.

for most of the Cretaceous [2], we infer that southwest Japan underwent a southward migration relative to Eurasia, with displacement of $18 \pm 8°$ in paleolatitude for the mid- to Late Cretaceous. In other words, a comparison of the paleolatitude of the Goshonoura Group, which is $45 \pm 7°$N, with the present position referred to west-central Kyushu (32°N), leads to the conclusion that proto-southwest Japan was located further to the north in excess of 10° in the mid-Cretaceous.

Finally, we interpret the southward migration of southwest Japan in terms of regional tectonics in the periphery of northeastern Asia to the east of the Siberian craton. Although the tectonic evolution of this entire area is still controversial, some authors regard this area as a collage of terranes or microplates of various ages and origins [36–38]. Paleomagnetic data are scarce, but a few studies by Russian workers [39–41] suggest that Mesozoic fragments of island-arc affinity originated far south from the present continental margin. We assume that these allochthonous complexes drifted northward, in association with the other exotic terranes like the Okhotsk Sea microplate, and they collided and accreted in succession with northeast Asia in the late Mesozoic. Accordingly, we postulate that proto-southwest Japan, after its amalgamation with northeast Asia, was transported southward along the continental margin to accommodate the penetration of such exotic blocks as the Okhotsk Sea microplate. We thus propose a possible tectonic model (Fig. 13), which is similar to those proposed for the India–Asia collision in that large-scale transport of intra-continental terranes occurs in the vicinity of a colliding landmass.

In our model, the Cretaceous pull-apart basins in East Asia including the Goshonoura basin lie along strike-slip faults, which were presumably reactivated by the collision of the Okhotsk Sea microplate. Previous geodynamic models of East Asia based on the Mesozoic plate reconstruction [5] suggest that sinistral displacement was dominant for the strike-slip fault systems in this region, in which exotic terranes underwent significant coast-parallel translation. However, we emphasize that the collision of the Okhotsk Sea microplate gave rise to a change in motion of the fault system, whereby pull-apart basins in the proto-Japanese arc migrated to the south. We propose that the deformation of the eastern margin of Asia would have been affected by suspected collision of allochthonous terranes over the mid- to Late Cretaceous time.

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